Hybrid finite element and neural network methods for parametric PDEs

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MACS seminar
("Modélisation, Analyse et Calcul Scientifique")

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Context and motivation

Many-query contexts: real-time simulation, digital twins, optimization, uncertainty quantification

$\Rightarrow$ **Goal:** solve a parametric PDE accurately and quickly, for many parameters $\boldsymbol{\mu}$

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Our focus: the expressivity of the approximation spaces

Two hybridizations

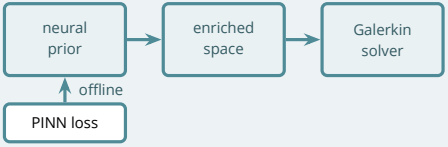
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1st hybridization

Enriched FEM using PINNs



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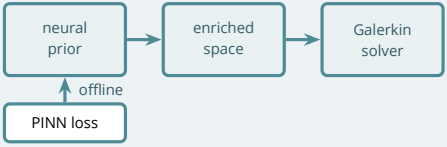
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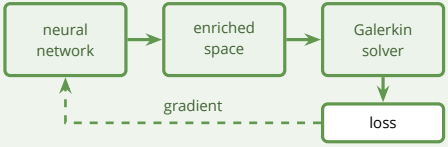
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2nd hybridization

Learnable Galerkin methods



Network: learned through the solver

Guarantees: none (experimental)

Results: 1D, 3D, parametric

Parametric elliptic PDEs

Let $\Omega \subset \mathbb{R}^d$ be the spatial domain ($d = 1, 2, 3$) and $\mathcal{M} \subset \mathbb{R}^p$ the parameter space.
For one or several $\boldsymbol{\mu} \in \mathcal{M}$, find $u(\,\cdot\,;\boldsymbol{\mu}) : \Omega \to \mathbb{R}$ such that

$$\begin{cases} \mathcal{L}\big(u\,;\boldsymbol{x},\boldsymbol{\mu}\big) = f(\boldsymbol{x},\boldsymbol{\mu}), & (\boldsymbol{x},\boldsymbol{\mu}) \in \Omega \times \mathcal{M}, \\ \text{Dirichlet, Neumann or Robin BC}, & (\boldsymbol{x},\boldsymbol{\mu}) \in \Gamma \times \mathcal{M}, \end{cases} \tag{$\mathcal{P}$}$$

with $\Gamma := \partial\Omega$ the boundary of Ω, f the source term and $\mathcal{L}$ the linear differential operator defined by

$$\mathcal{L}(\cdot\,;\boldsymbol{x},\boldsymbol{\mu}) : u \mapsto R(\boldsymbol{x},\boldsymbol{\mu})\,u + \boldsymbol{C}(\boldsymbol{x},\boldsymbol{\mu}) \cdot \nabla u - \frac{1}{\mathrm{Pe}} \nabla \cdot \big(D(\boldsymbol{x},\boldsymbol{\mu}) \nabla u\big).$$

where R is the reaction, $\boldsymbol{C}$ the convection, D the (symmetric positive definite) diffusion matrix and Pe the Péclet number.

Numerical methods

Numerical methods for parametric PDEs*

* presented with homogeneous Dirichlet boundary conditions

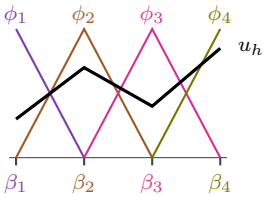
Continuous Galerkin (CG) method[1]

For a fixed $\boldsymbol{\mu} \in \mathcal{M}$, let $\mathcal{T}_h$ be a mesh of Ω.

Let V_h be a continuous space of piecewise polynomials of degree k on $\mathcal{T}_h$. Its basis functions $(\phi_i)_i$, with ϕ_i supported on the cells sharing the dof i, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{i=1}^{N_{\text{dofs}}} \beta_i(\boldsymbol{\mu})\, \phi_i(\boldsymbol{x}).$$

$N_{\text{cells}} = 3$, $N_{\text{dofs}} = 4$ $(k = 1)$



[1] [Ern and Guermond, 2004]

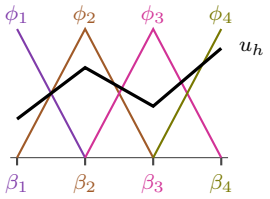
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with $V_h^0 \subset V_h$ vanishing on Γ (homogeneous Dirichlet condition).

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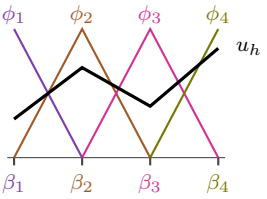
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Linear system:

$$\underset{\text{matrix}}{A_h(\boldsymbol{\mu})}\ \underset{\text{dofs}}{\boldsymbol{\beta}_h(\boldsymbol{\mu})} = \underset{\text{right-hand side}}{\boldsymbol{b}_h(\boldsymbol{\mu})}, \qquad A_{ij} = a(\phi_j, \phi_i; \boldsymbol{\mu}), \quad b_i = l(\phi_i; \boldsymbol{\mu})$$

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Physics-informed neural networks (PINNs)[1]

Let $u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$ be a neural network with weights θ and inputs $(\boldsymbol{x}, \boldsymbol{\mu}) \in \Omega \times \mathcal{M}$, approximating the solution u.

Training: Find the weights $\theta^\star$ such that

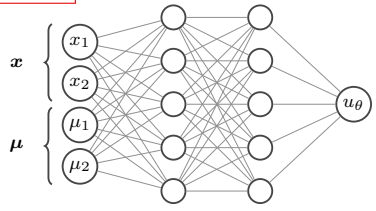
$$\theta^\star = \arg\min_\theta \big(\omega_r \, \mathcal{J}_r(\theta) + \omega_b \, \mathcal{J}_b(\theta)\big), \qquad (\mathcal{P}_\theta)$$

with ω_r and ω_b the weights of the two loss terms, and

residual loss
$$\mathcal{J}_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} \big|\mathcal{L}\big(u_\theta(\boldsymbol{x}, \boldsymbol{\mu}); \boldsymbol{x}, \boldsymbol{\mu}\big) - f(\boldsymbol{x}, \boldsymbol{\mu})\big|^2 \, \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu},$$

boundary loss
$$\mathcal{J}_b(\theta) = \int_{\mathcal{M}} \int_{\Gamma} \big|u_\theta(\boldsymbol{x}, \boldsymbol{\mu})\big|^2 \, \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu}.$$

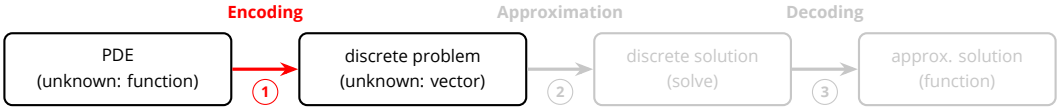
Monte-Carlo method:
Discretize the cost functions by random process.



[1] [Raissi, Perdikaris, and Karniadakis, 2019]

Link between Galerkin and PINNs

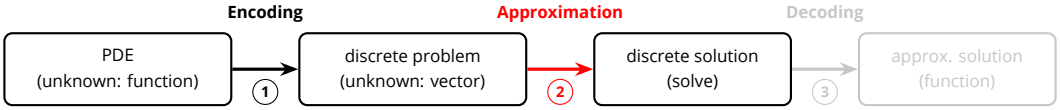
Numerical method (three steps):



		Galerkin (CG)	PINN
	Space	vector space V_h	manifold of the u_θ
①	Unknowns	dofs $\boldsymbol{\beta}_h(\boldsymbol{\mu})$	weights θ
	Integrals	quadrature on cells	Monte Carlo on collocation points

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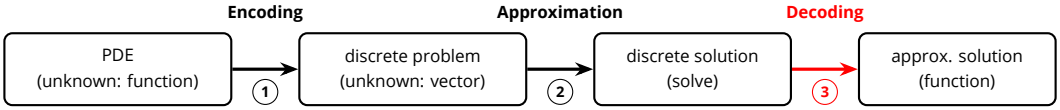
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	Resolution	linear system $A_h(\boldsymbol{\mu})\boldsymbol{\beta}_h(\boldsymbol{\mu}) = \boldsymbol{b}_h(\boldsymbol{\mu})$	optimization on θ

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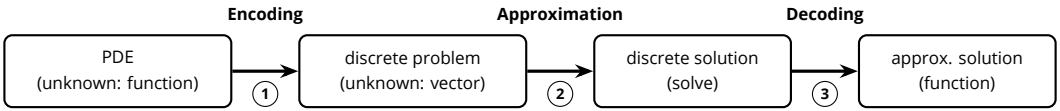
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Link between Galerkin and PINNs

Numerical method (three steps): **Same philosophy, different approximation space**



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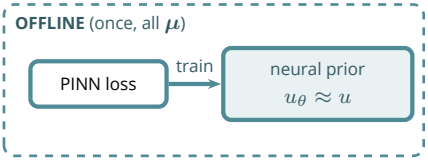
1st hybridization

Enriched finite element method using PINNs

Presentation*

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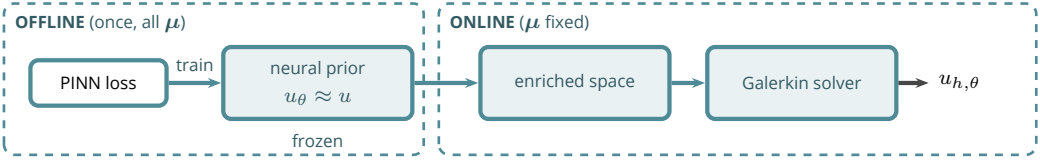
Pipeline of the hybrid method[1]



Offline: train a parametric PINN $u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$ once, for all $\boldsymbol{\mu} \in \mathcal{M}$.

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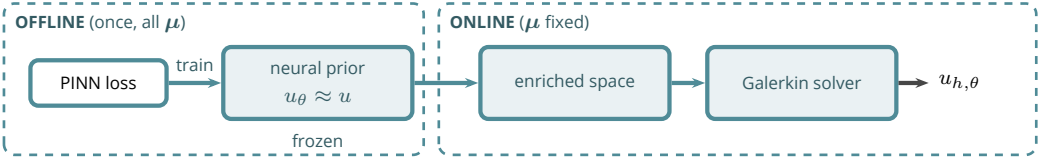


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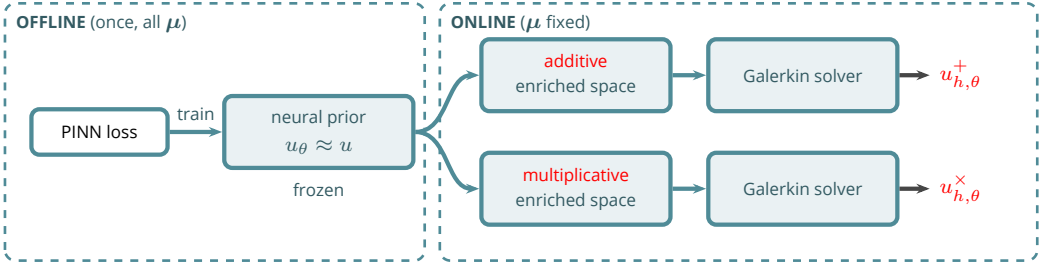
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Goal: accuracy of a fine mesh on a coarser mesh.

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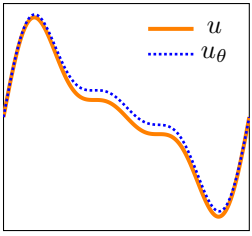
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Two versions of $u_{h,\theta}$: additive $u^+_{h,\theta}$ and multiplicative $u^\times_{h,\theta}$.

[1] [Lecourtier et al., 2026]

Additive approach[1]

Prior: Let $u_\theta \in H^{k+1}(\Omega)$ be the PINN prediction ($\boldsymbol{\mu}$ fixed).



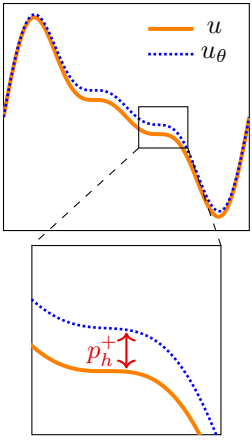
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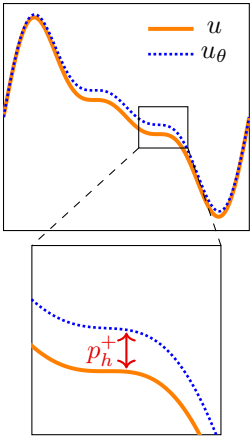
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with modified Dirichlet BC. Appendix A1



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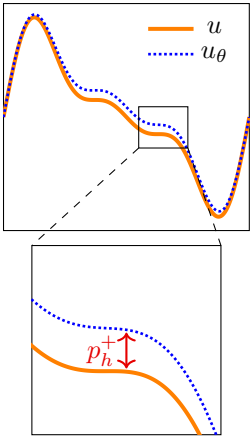
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$$\|u - u_h\|_{L^2} \leqslant C_{L^2}\, h^{k+1} |u|_{H^{k+1}}.$$

[1] Stated in L^2 norm; similar result in H^1 semi-norm.

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Theorem 2: Additive approach [Lecourtier et al., 2026]

Let $u_{h,\theta}^+ \in V_h^+$ be the solution of $(\mathcal{P}_h^+)$ with V_h^+ the enriched trial space. Then,

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with $C_{\theta,M} = \|u_{\theta,M}^{-1}\|_{L^\infty} + 2|u_{\theta,M}^{-1}|_{W^{1,\infty}} + |u_{\theta,M}^{-1}|_{W^{2,\infty}}$

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The multiplicative gain converges to the additive gain when $M \to \infty$:

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Numerical results

1D Poisson
2D Poisson
3D Poisson

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1D Poisson
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Problem considered

TODO

Online phase - 1 parameters instance I

TODO

Online phase - 1 parameters instance II

TODO

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Parametric framework

TODO

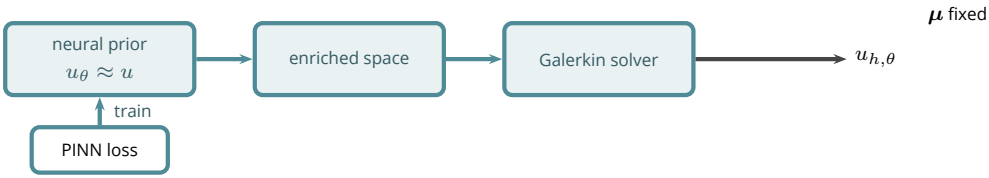
2nd hybridization

Learnable Galerkin methods

Introduction

From frozen prior to learned basis

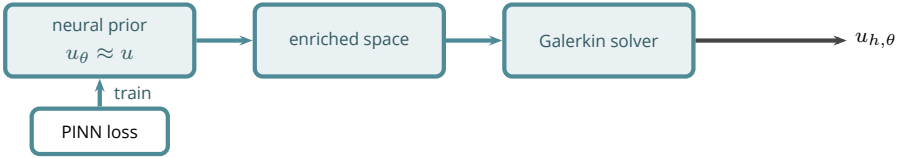
1st hybridization: neural prior trained offline, then frozen: a priori error estimates.
Limit: the gain is bounded by the quality of the prior, trained apart from the scheme.



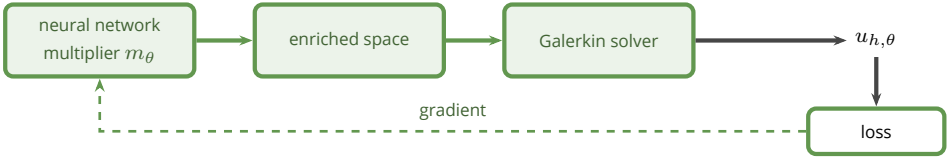
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μ fixed



2nd hybridization: learn the basis through the solver, adapted to the solution of the scheme.
Advantage: basis adapted to the scheme, not bounded by a frozen prior.
Limit: no a priori guarantees, training cost.



Automatic differentiation and ScimBa

Need: gradient of the solver w.r.t. the N_θ weights, without deriving it by hand.

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Three ways to compute the gradient:

	exact	use	cost per gradient
symbolic	✓	closed form only	huge expressions
finite differences	×	any program	N_θ solves
automatic diff.[1]	✓	any program	a few solves

[1] [Baydin et al., 2018]

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ScimBa: the whole pipeline is a JAX program[2]:

$$\text{network} \to \text{basis} \to \text{assembly} \to \text{solve} \to \text{loss } \mathcal{J}$$

One backward pass gives $\partial \mathcal{J} / \partial \theta$ through all the steps.

See ScimBa

[1] [Baydin et al., 2018]
[2] [Bradbury et al., 2018]

Non-parametric enrichment (PINN loss)*

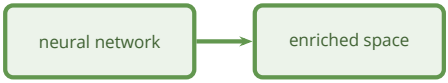
* μ is fixed and omitted from the notation: basis, dofs and solution are valid for this μ only

Neural-enriched Galerkin scheme

neural network

Neural network: Construct a single neural network $v_\theta(\boldsymbol{x})$ on Ω (with N_θ weights), shared by all cells.

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Galerkin scheme (μ fixed): Solve the linear system:

$$\boldsymbol{R}_h\big(\boldsymbol{\beta}_h(\theta);\theta\big) = A_h(\theta)\,\boldsymbol{\beta}_h(\theta) - \boldsymbol{b}_h(\theta) = \boldsymbol{0},$$

with $\quad A_{(c,i),(c',j)}(\theta) = a_h^{\text{disc}}(\phi^\theta_{c',j}, \phi^\theta_{c,i}), \qquad b_{(c,i)}(\theta) = l(\phi^\theta_{c,i}).$

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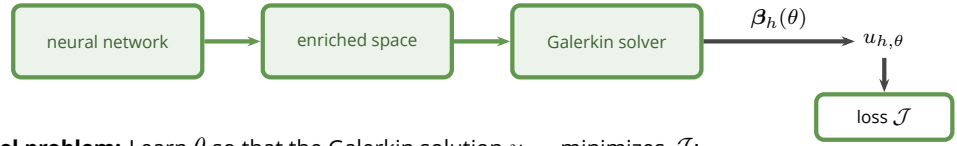
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Solution: Reconstruct the solution in the enriched space as:

$$u_{h,\theta}(\boldsymbol{x}) = \sum_{c,i} \beta_{c,i}(\theta)\,\phi^\theta_{c,i}(\boldsymbol{x}).$$

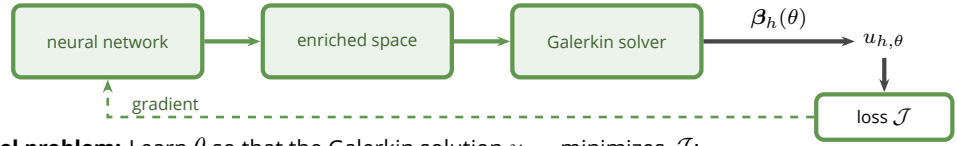
Learning θ: differentiating through the solver



Bi-level problem: Learn θ so that the Galerkin solution $u_{h,\theta}$ minimizes $\mathcal{J}$:

$$\theta^\star = \arg\min_\theta \ \mathcal{J}(\boldsymbol{\beta}_h(\theta);\theta) \quad \text{s.t.} \quad \boldsymbol{R}_h(\boldsymbol{\beta}_h(\theta);\theta) = \mathbf{0}$$

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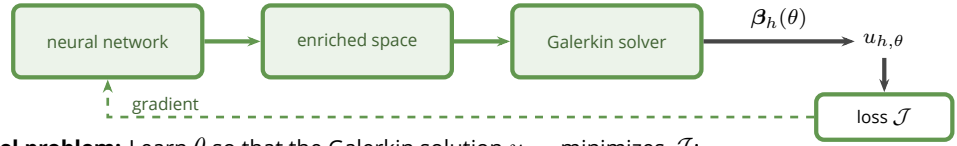
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Adjoint: With a gradient-based optimizer, avoid $d\boldsymbol{\beta}_h/d\theta$ (only one linear solve, whatever N_θ):

$$A_h(\theta)^T \boldsymbol{\lambda}_h = \left(\frac{\partial \mathcal{J}}{\partial \boldsymbol{\beta}_h}\right)^T \ \Rightarrow \ \frac{d\mathcal{J}}{d\theta} = \left.\frac{\partial \mathcal{J}}{\partial \theta}\right|_{\boldsymbol{\beta}_h} - \boldsymbol{\lambda}_h^T \frac{\partial \boldsymbol{R}_h}{\partial \theta}(\boldsymbol{\beta}_h;\theta)$$

1D test case and PINN loss

Test case: 1D advection-diffusion with a boundary layer, $\Omega = (0,1)$, $\mathrm{Pe} = 20$, $f = 1$:

$$\begin{cases} -\dfrac{1}{\mathrm{Pe}}\,\partial_{xx}u + \boldsymbol{C}\,\partial_x u = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma, \end{cases} \qquad \boldsymbol{C}(x,\boldsymbol{\mu}) = \mu_1 + \mu_2 x$$

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$$\text{new } \boldsymbol{\mu},\ k \text{ or } N_{\text{cells}} \ \Rightarrow\ \text{new training.}$$

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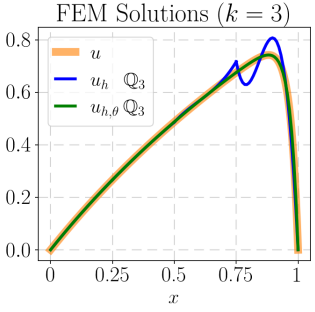
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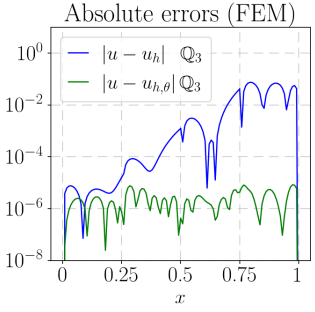
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Evaluation: Relative L^2 error w.r.t. a reference FEM solution (128 cells, $\mathbb{Q}_4$), never used in the training.

Learned basis: affine multiplier

Setting: CG, $N_{\text{cells}} = 4$ cells, $\mathbb{Q}_3$ ($N_{\text{dofs}} = 13$), affine multiplier $m_\theta = 2 + 0.1\, v_\theta$.
Training: MLP v_θ (tanh, 2×16, $N_\theta = 321$), ENG optimizer[1], 300 epochs, 6000 collocation points.



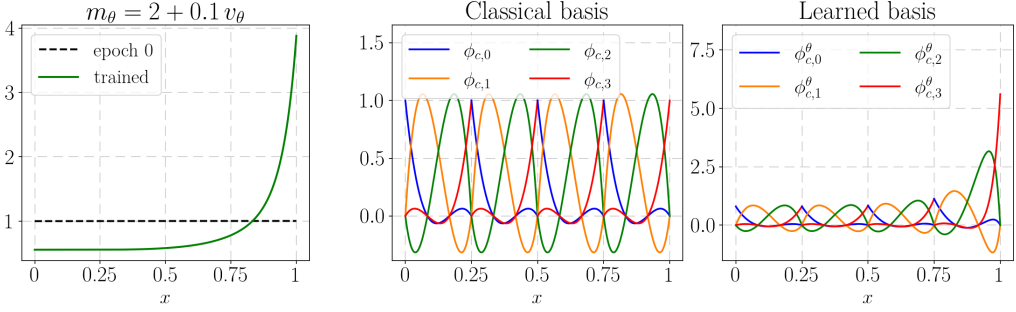


Solutions: Classical u_h oscillates in the boundary layer, $u_{h,\theta}$ matches the reference ($\sim 10^{-5}$ against $7 \cdot 10^{-2}$).

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Basis: The multiplier grows in the boundary layer: the last basis functions are amplified.

[1] [Müller and Zeinhofer, 2023]

Choice of the multiplier

Fix $N_{\text{dofs}} = 25$ and $N_\theta \in \{134, \dots, 151\}$, vary the levers: (N_{cells}, k) and the MLP shape (width, depth).

20 configurations $\times$ 20 seeds $=$ 400 trainings per multiplier (100 epochs, 5000 collocation points).

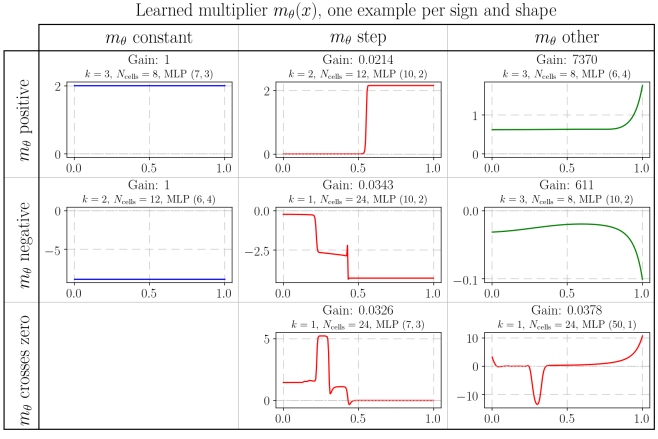
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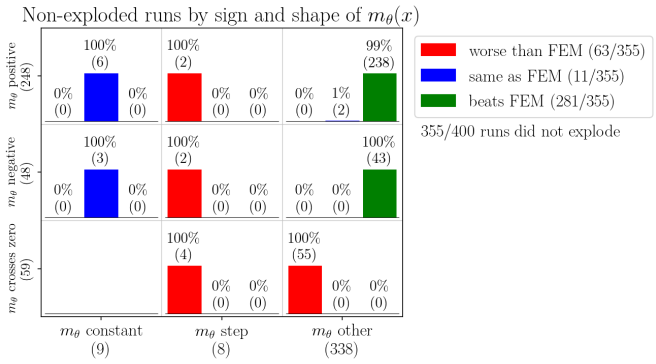
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Steps and sign changes never beat FEM.



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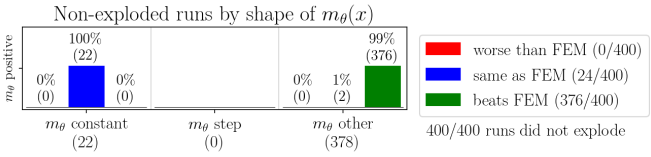
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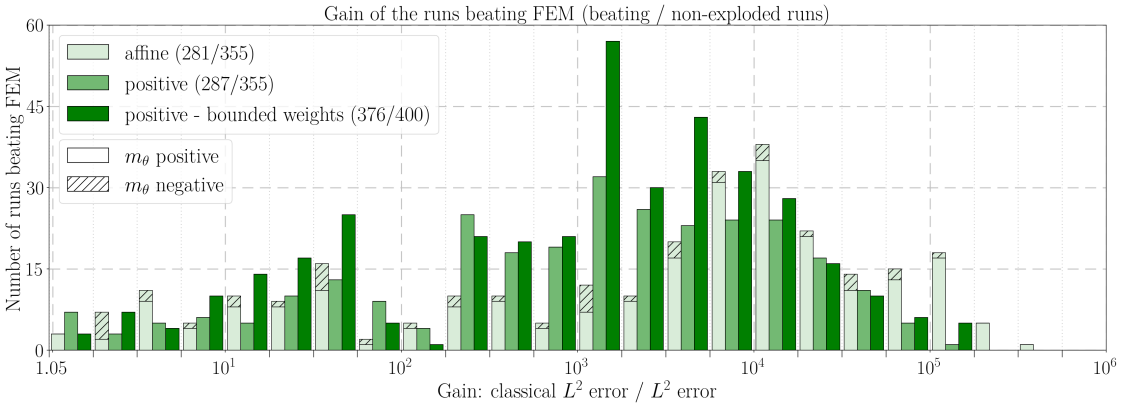
Fix: Positive multiplier

$$m_\theta = \exp(2 \tanh v_\theta)$$

with bounded weights:

No sign change, no step.

Results at fixed budget



Gain: $376/400$ beat the FEM, none explodes; the affine multiplier keeps the highest gains.

Results at fixed budget

(N_{cells}, k) / N_{dofs} (FEM discretization)	Relative L^2 error classical FEM	Median relative L^2 error (best one) over the seeds reaching the loss threshold (best loss $< 10^{-3}$)			
		(width, depth) N_θ: (50, 1) 151	(10, 2) 141	(7, 3) 134	(6, 4) 145
(24, 1) 25	$2.5 \cdot 10^{-2}$	threshold not reached	threshold not reached	threshold not reached	threshold not reached
(12, 2) 25	$1.6 \cdot 10^{-2}$	$4.3 \cdot 10^{-5}$ ($2.5 \cdot 10^{-5}$)	$2.5 \cdot 10^{-6}$ ($8.5 \cdot 10^{-7}$)	$6.9 \cdot 10^{-7}$ ($1.6 \cdot 10^{-7}$)	$3.8 \cdot 10^{-7}$ ($9.9 \cdot 10^{-8}$)
(8, 3) 25	$8.6 \cdot 10^{-3}$	$6.6 \cdot 10^{-6}$ ($5.0 \cdot 10^{-6}$)	$1.7 \cdot 10^{-6}$ ($3.2 \cdot 10^{-7}$)	$2.1 \cdot 10^{-6}$ ($5.9 \cdot 10^{-8}$)	$1.2 \cdot 10^{-6}$ ($1.9 \cdot 10^{-7}$)
(6, 4) 25	$4.7 \cdot 10^{-3}$	$3.3 \cdot 10^{-6}$ ($1.2 \cdot 10^{-6}$)	$1.1 \cdot 10^{-6}$ ($3.6 \cdot 10^{-7}$)	$9.0 \cdot 10^{-7}$ ($2.7 \cdot 10^{-7}$)	$1.0 \cdot 10^{-6}$ ($2.5 \cdot 10^{-7}$)
(4, 6) 25	$1.3 \cdot 10^{-3}$	$3.8 \cdot 10^{-6}$ ($1.0 \cdot 10^{-6}$)	$6.3 \cdot 10^{-7}$ ($1.1 \cdot 10^{-7}$)	$2.7 \cdot 10^{-7}$ ($5.7 \cdot 10^{-8}$)	$3.6 \cdot 10^{-7}$ ($2.6 \cdot 10^{-8}$)

FEM discretization: N_{cells} ↘, k ↗ — MLP shape: width →, depth ↗

Gain: $376/400$ beat the FEM, none explodes; the affine multiplier keeps the highest gains.

Levers: degree and depth matter: $k = 1$ only a modest gain over FEM, $(50, 1)$ is the worst network.

3D test case

Test case: 3D advection-diffusion with a boundary layer, $\Omega = (0,1)^3$, $\mathrm{Pe} = 20$:

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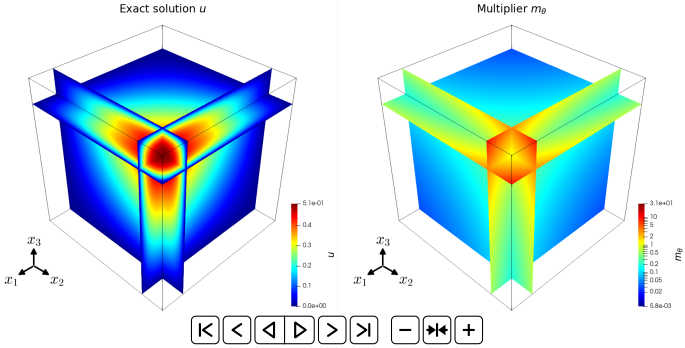
Evaluation: Relative L^2 error w.r.t. the manufactured solution u, never used in the training.

3D: learned multiplier

Setting: CG, $N_{\text{cells}} = 4 \times 4 \times 4$ cells, $\mathbb{Q}_2$ ($N_{\text{dofs}} = 729$), positive multiplier $m_\theta = \exp\left(6 \tanh v_\theta\right)$.
Training: MLP v_θ (tanh, 3×10, $N_\theta = 271$), ENG optimizer, 70 epochs, 6000 collocation points, 3 seeds.

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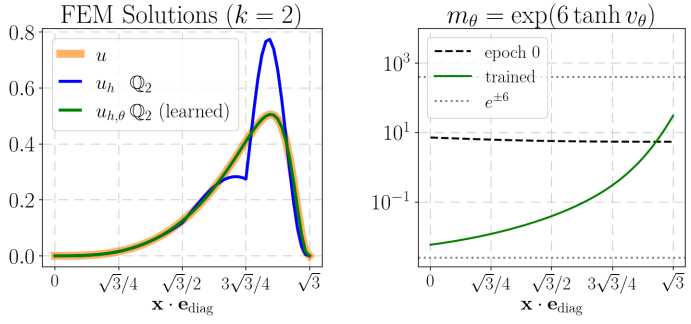
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Solution: classical u_h overshoots, $u_{h,\theta}$ matches u.

3D: gain and cost

	time (s)									
	w/ comp.		w/o comp.		N_{dofs}			error		
seed	learn.	class.	learn.	class.	learn.	class.	ratio	learn.	class.	ratio
1	59	51	10.1	49.5	729	$5.31 \cdot 10^5$	729	$5.68 \cdot 10^{-5}$	$3.53 \cdot 10^{-4}$	6.2
2	41	37	6.8	26.6	729	$3.89 \cdot 10^5$	534	$8.13 \cdot 10^{-5}$	$4.82 \cdot 10^{-4}$	5.9
3	41	37	6.9	26.6	729	$3.89 \cdot 10^5$	534	$4.77 \cdot 10^{-4}$	$4.82 \cdot 10^{-4}$	1

Equal time: error $\times 6$ lower for seeds $1, 2$, equal for seed 3.

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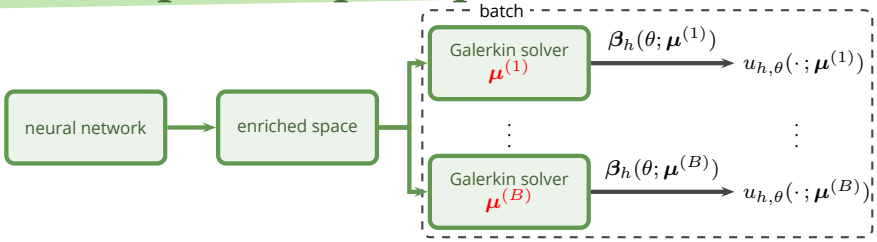
Dofs: 534 to 729 times fewer for the learned basis (without counting the $N_\theta = 271$ weights).

Cost: learned time dominated by compilation: 7 to 10 s without it, against 27 to 50 s for the FEM.

Parametric enrichment (neural operator)*

* $\boldsymbol{\mu}$ is no longer fixed: one network trained once, for all $\boldsymbol{\mu} \in \mathcal{M}$

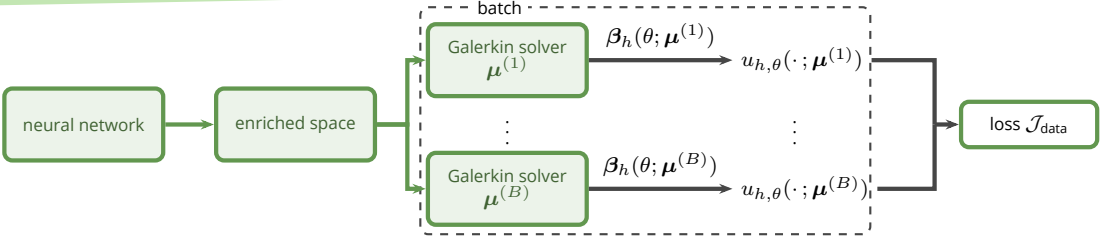
Neural operator: principle



Parameter sampling: $\mathcal{D}_{\text{train}} \subset \mathcal{M}$ chosen once, a random batch of $\boldsymbol{\mu}$ drawn at each epoch.

Parametric scheme: One network, one Galerkin solve per $\boldsymbol{\mu}$ of the batch: $A_h(\theta;\boldsymbol{\mu})\,\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}) = \boldsymbol{b}_h(\theta;\boldsymbol{\mu})$.

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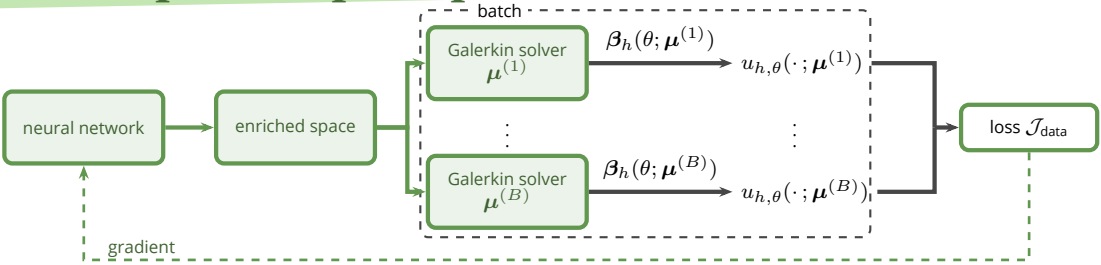
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Supervised loss: Fit reference solutions on the batch, at N_x points per $\boldsymbol{\mu}$:

$$\mathcal{J}_{\text{data}}(\theta) = \frac{1}{B\,N_x}\sum_{i=1}^{B}\sum_{j=1}^{N_x}\Big|u_{h,\theta}\big(\boldsymbol{x}^{(j)};\boldsymbol{\mu}^{(i)}\big) - u_{\text{ref}}\big(\boldsymbol{x}^{(j)};\boldsymbol{\mu}^{(i)}\big)\Big|^2.$$

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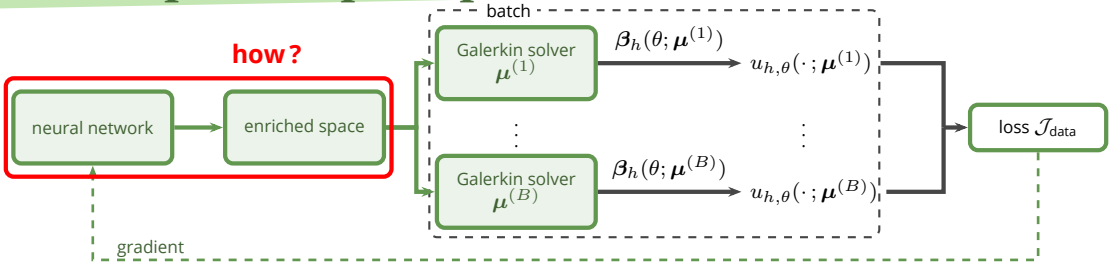
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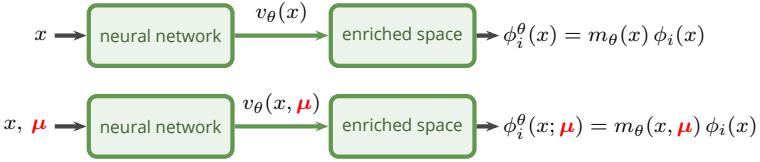
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Constructing the enriched space (three ways)



Shared basis: same enriched space for all μ.

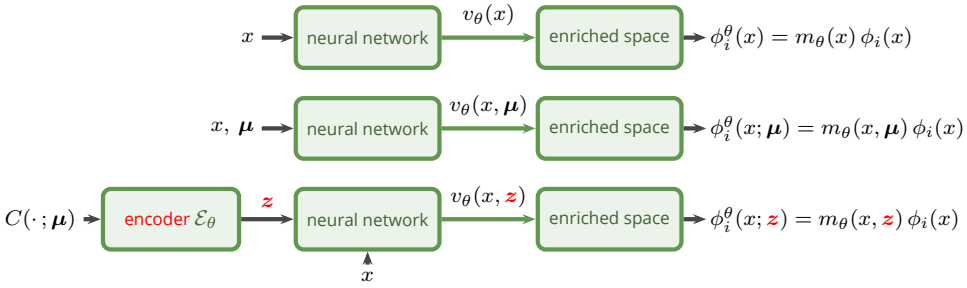
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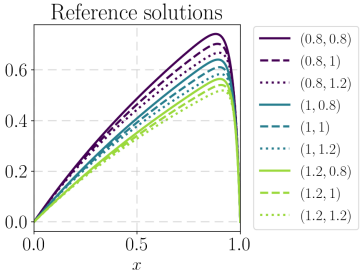
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Field-dependent basis: an encoder summarizes the transport field in a latent state $\boldsymbol{z}$.

1D linear transport

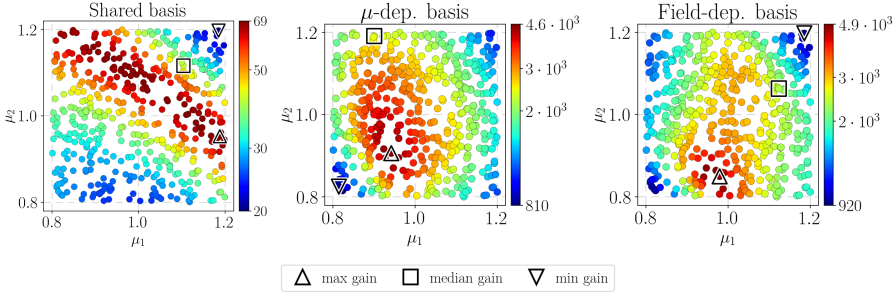
Test case: Same 1D case as the PINN, with $\boldsymbol{C}(x, \boldsymbol{\mu}) = \mu_1 + \mu_2 x$, $\boldsymbol{\mu} \in [0.8, 1.2]^2$: the layer stays at $x = 1$.



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Setup: DG, $N_{\text{cells}} = 8$, $k = 2$, 32 training $\boldsymbol{\mu}$, batch of 16, $N_x = 101$ points per $\boldsymbol{\mu}$.

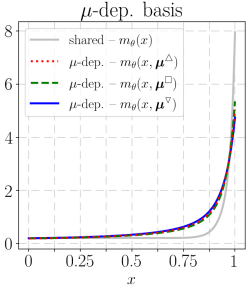


Gain (on 500 unseen $\boldsymbol{\mu}$):
shared 20–69 ($N_\theta = 321$), $\boldsymbol{\mu}$-dep. 811–$4.6 \cdot 10^3$ ($N_\theta = 353$), field-dep. 917–$4.9 \cdot 10^3$ ($N_\theta = 1381$).

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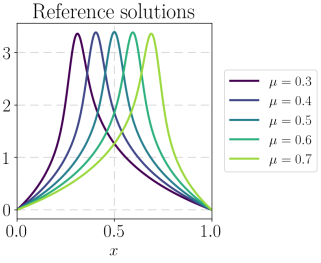


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Multipliers: shared, rises only in the last cells; $\boldsymbol{\mu}$-dep., rises over more cells.

1D turning point

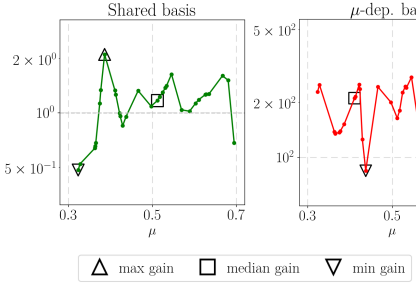
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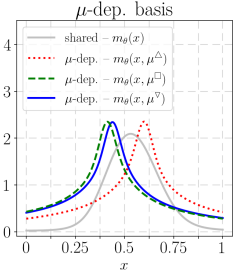


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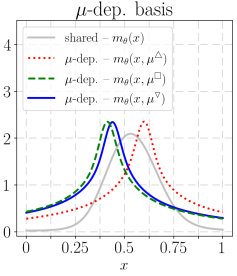
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Conclusion

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1st hybridization: enriched FEM with a frozen PINN prior

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2nd hybridization: learnable Galerkin bases

Limits and perspectives

TODO

Appendix – 1st hybridization

Enriched finite element method using PINNs

Additional details

General Dirichlet boundary conditions

Problem: $u = g$ on Γ, with u_θ the PINN prior.

Additive approach:

$$p_h^+ = g - u_\theta \quad \text{on } \Gamma.$$

Homogeneous as soon as u_θ is exact on Γ (exact BC in the PINN).

Multiplicative approach ($u_{h,\theta}^\times = u_{\theta,M}\, p_h^\times - M$, $u_{\theta,M} = u_\theta + M$):

$$\text{standard PINN:} \quad p_h^\times = \frac{g + M}{u_{\theta,M}} \quad \text{on } \Gamma,$$

$$\text{PINN with exact BC:} \quad p_h^\times = 1 \quad \text{on } \Gamma.$$

Homogeneous case: $M = 0$ possible, $p_h^\times$ left free (more freedom for the derivatives at the boundary).

Curved boundaries (disk, $k > 2$): the prior is not exact on $\partial\Omega_h$; in the tests, $g = u$ on $\partial\Omega_h$.

Appendix – 2nd hybridization

Learnable Galerkin methods

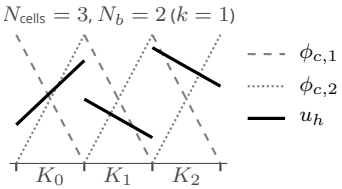
Additional details

Discontinuous Galerkin (DG) method[1]

For a fixed $\boldsymbol{\mu} \in \mathcal{M}$, let $\mathcal{T}_h$ be a mesh of Ω of cells $(K_c)_{0 \leq c < N_{\text{cells}}}$.

Let V_h^{disc} be a discontinuous space of piecewise polynomials of degree k on the mesh $\mathcal{T}_h$. Its basis functions $(\phi_{c,i})_{c,i}$, with $\phi_{c,i}$ supported on a single cell K_c, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{c,i} \beta_{c,i}(\boldsymbol{\mu})\, \phi_{c,i}(\boldsymbol{x}).$$



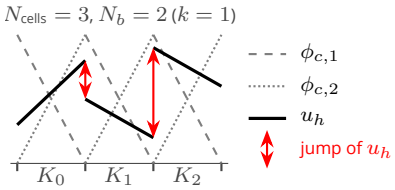
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Coupling the cells: ambiguous traces on the interfaces $\Rightarrow$ numerical flux (it defines the method)

Diffusion-dominated regimes: flux for the diffusion term only, here SIPG (Symmetric Interior Penalty Galerkin)

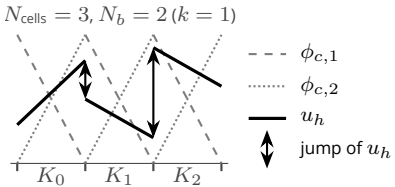
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Variational problem (SIPG):

Find $u_h \in V_h^{\text{disc}}$ such that, $\forall v_h \in V_h^{\text{disc}}$, $a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = l(v_h; \boldsymbol{\mu}) = \int_\Omega f\, v_h$, with

$$a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = \underbrace{\int_\Omega \left(R\, u_h v_h + \boldsymbol{C} \cdot \nabla_h u_h\, v_h + \frac{1}{\text{Pe}} D \nabla_h u_h \cdot \nabla_h v_h \right)}_{\text{cell terms}} \underbrace{- \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \left([\![u_h]\!] \cdot \{D \nabla_h v_h\} + \{D \nabla_h u_h\} \cdot [\![v_h]\!] \right)}_{\text{consistency}} \underbrace{+ \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \kappa\, [\![u_h]\!] \cdot [\![v_h]\!]}_{\text{penalty}}$$

where $\mathcal{F}_h$ is the set of faces, ∇_h the broken gradient, $[\![\cdot]\!]$ the jump, $\{\cdot\}$ the average, κ the penalty parameter.

[1] [Arnold et al., 2002]

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

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TODO

Résultats numériques supplémentaires